

The effect of discourse structure on depth of semantic integration in reading

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Abstract A coherent discourse exhibits certain structures in that subunits of discourses are related to one another in various ways and in that subunits that contribute to the same discourse purpose are joined to create a larger unit so as to produce an effect on the reader. To date, this crucial aspect of discourse has been largely neglected in the psycholinguistic literature. In two experiments, we examined whether semantic integration in discourse context was influenced by the difference of discourse structure. Readers read discourses in which the last sentence was locally congruent but either semantically congruent or incongruent when interpreted with the preceding sentence. Furthermore, the last sentence was either in the same discourse unit or not in the same discourse unit as the preceding sentence, depending on whether they shared the same discourse purpose. Results from self-paced reading (Experiment 1) and eye tracking (Experiment 2) showed that discourse-incongruous words were read longer than discourse-congruous words only when the critical sentence and the preceding sentence were in the same discourse unit, but not when they belonged to different discourse units. These results establish discourse structure as a new factor in semantic integration and suggest that discourse effects depend both on the content of what is being said and on the way that the contents are organized.

Keywords Semantic integration · Discourse structure · Within-unit structure · Between-unit structure

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An important process during natural language comprehension is the establishment of meaning in a discourse context. In order to understand natural discourse comprehension, it is necessary to know how the comprehender integrates local information into the preceding discourse context and what factors affect this integration process. A substantial body of literature has been devoted to the investigation of how readers interpret upcoming information with the preceding discourse context. However, linguistic studies have long suggested that discourse effects involve not just what the preceding sentences were about in the discourse context, but also how the discourse is organized. In the present study, we examine the role that discourse structure has in influencing semantic integration.

Semantic integration in discourse context

A number of ERP studies have examined how local information is related to general discourse context (Camblin et al. 2007; Filik and Leuthold 2008; George et al. 1999; Hald et al. 2007; Nieuwland and Van Berkum 2005, 2006; Nieuwland, Van Salmon, and Pratt 2002; Van Berkum et al. 1999, 2003, 2005). Conflicts between discourse context and local information result in an N400 effect for inappropriate words, as opposed to appropriate words. The N400 is a centro-parietal negativity, peaking around 400 ms post-stimulus-onset. The amplitude of N400 is sensitive to the ease with which words are integrated into the previous context. It was first reported by Kutas and Hillyard (1980) for semantic anomalies in sentence context. Generally, semantically incongruous words elicit larger N400s than do semantically congruous words. This is referred to as an N400 effect (for a review, see Kutas and Federmeier 2000). Van Berkum et al. (1999), for example, investigated discourse- and sentence-dependent anomalies in two ERP experiments. In Experiment 1, participants were presented with short stories. The last sentence of the stories

contained a critical word that was always coherent in the local sentence context but was either discourse congruous or discourse incongruous. It was found that as compared with discourse-congruous words, discourse anomalies elicited a large N400 effect. In Experiment 2, the same sentences were presented in isolation. The results showed that the N400 effect found in Experiment 1 was much reduced. In addition, comparison between sentence-dependent anomalies and discourse-dependent anomalies showed that the two anomalies shared a similar time course, overall morphology, and scalp distribution. This forcefully illustrates the fact that words of an unfolding sentence are immediately mapped into the wider discourse context.

Eye movement studies paint a similar picture. The ability of context to influence the interpretation of local information has been demonstrated repeatedly (Camblin et al. 2007; Filik 2008; Warren et al. 2008). An experiment conducted by Filik demonstrates that by placing a pragmatically anomalous sentence (such as “The mouse picked up the dynamite”) in a fictional scenario that is already well known to the reader (e.g., a Tom and Jerry cartoon), the difficulty usually associated with these pragmatic anomalies can be eliminated, as reflected in the first-fixation duration and first-pass reading times in the posttarget region. A similar study conducted by Warren et al. tested sentences like “Harry used a microwave to heat/teach the tough bread” in either a real-world context or a fantasy context. Gaze durations on a target word were longer to impossible events independently of context. However, a measure of the time elapsed from first fixating the target word to moving past it showed disruption only in the real-world context. These results appear to conflict with an immediate effect of context as observed in the ERP studies and suggest a delayed effect of context in which contextual information did not eliminate initial disruption but moderated it quickly thereafter. However, the discrepancy with regard to how quickly the effect of context takes place may largely be due to the difference in contextual constraint. When contextual constraints are weak, initial processing is dominated by local lexical-semantic factors, with effects of context emerging later (for discussions, see Nieuwland 2013; Nieuwland and Martin 2012).

Note that discourse context may not always override local anomaly. When contextual support for discourse anomalies is very strong, the anomalies can go undetected. Prior literature has observed that difficult anomalies having a good fit to general context did not produce an N400 effect, while control “easy-to-detect” anomalies did. This is interpreted as evidence of shallow processing in which the intended message seems to be understood before a complete semantic analysis has been carried out (Sanford et al. 2011). Shallow processing can also happen when local anomalies receive incomplete processing due to low attention allocation or processing difficulty (Barton and Sanford 1993; Sanford and Graesser 2006; Sanford and Sturt 2002; Ward and Sturt 2007). In contrast to shallow

processing, semantic information can also be processed quite elaborately in discourse context. There are studies showing that semantic anomalies were processed more elaborately when marked with pitch accent or when they were in focus position (Wang et al. 2011; Wang et al. 2009). For instance, in Wang et al. (2009), anomalous words in focus position elicited a full N400 effect, while those in nonfocus position elicited a strongly reduced N400 effect. Thus, semantic information can be processed either more or less elaborately depending on linguistic context. Furthermore, depth of semantic processing may reflect the extent to which local information is interpreted with the preceding context (Sanford and Graesser 2006; Sanford et al. 2011; Sanford and Sturt 2002).

The robust effect of discourse-dependent anomalies clearly shows that readers and listeners immediately relate an upcoming word to the preceding discourse context. However, the contribution from discourse structure was neglected in previous studies. Linguistic studies have long suggested that discourse structure is relevant to the interpretation of local information (den Ouden et al. 2009; Grosz and Sidner 1986; Nakajima and Allen 1993; Noordman et al. 1999; Smith 2004; Swerts 1997; Swerts and Ostendorf 1997; Taboada 2006; Taboada and Mann 2006b). According to rhetorical structure theory (hereafter referred to as RST), a notable theory in discourse analysis, a coherent discourse exhibits a hierarchical and connected structure in that subunits of discourses are related to one another in various ways (den Ouden et al. 2009; Noordman et al. 1999; Taboada 2006; Taboada and Mann 2006a, 2006b; Yang and Yang 2012). However, up until now, no experiment has yet been dedicated to accessing the role of discourse structure in semantic integration.

RST and discourse structure

There is general agreement that subunits of discourses are related to one another in various ways. Formalizing and understanding how the contents of a discourse are organized are fundamental for understanding a discourse, but psycholinguistic accounts of these phenomena are still limited. RST, developed by Mann and Thompson (1988), seeks to explain the coherence of discourse by describing how each individual unit of a discourse contributes to the overall goals of the discourse. Since its development, it has been successfully applied in diverse fields, such as computational linguistics, cross-linguistic studies, dialogue and multimedia, discourse analysis, argumentation, and writing (for a review, see Taboada and Mann 2006a, 2006b).

In the framework of RST, coherence of a discourse involves finding an intended role for every unit of the discourse. Two clauses in a discourse that share the same intended function are joined to create a new unit and to convey the purpose of the writer. This new unit may, in turn, be related to

another clause to create a new effect on the reader. This process can go on and on until, finally, the intended roles of every clause are organized in a hierarchical manner to produce the impression of coherence in discourse. In the discourse hierarchy, each clause is related to another clause or to a discourse unit consisting of two or more clauses.

For example, in discourse 1a (with translation below the Chinese sentences), the intended role of the first clause is to describe what the family said about Li, while both the second and third clauses play the function of describing what Li is up to recently. Therefore, the second and third clauses both answer the same subquestion (“What is Li up to recently?”), and they are subunits fulfilling the same discourse purpose. In terms of the effect on the reader, these two clauses should be joined to create a larger discourse unit before they are linked to the first clause, which obviously differs from these two clauses in the effect to be achieved on the reader (see Fig. 1a for the RST structure of discourse 1a). In discourse 1b, however, the first and second clauses describe what Li is up to recently, while the last clause describes what the family said about Li. Therefore, the first and second clauses are subunits for the same discourse purpose, and they create a larger discourse unit first before they are related to the last clause (see Fig. 1b for the RST structure of discourse 1b). Thus, although discourses 1a and 1b have roughly the same content, their structures differ in the way that the contents are organized.

1a家人都说小李准备参加运动会,近来他总去田径场,一遍遍地努力练习跑步准备比赛。The family says that Li is going to take part in the sports meeting. Lately, he's been going to the track field to practice running over and over again for the game.

1b最近小李周末总是出门,一遍遍地在田径场训练,家人都说他是去练习跑步准备比赛。

Lately, Li often goes out on weekends to train in the track field. His family says that he goes there to practice running for the game.

Of particular interest to us is that not all adjacent sentences in a discourse are closely related to each other. Two successive sentences may be in the same discourse unit (both are subunits of the same discourse purpose; see the second and third clauses in discourse 1a) or not in the same discourse unit (one is a subunit of a discourse purpose, while the other fulfills another discourse purpose; see the second and third clauses in discourse 1b), which results in a minor or a more major discourse boundary. It is worth noting that this representation

of discourse structure is not just following the intuitions of linguistic theoreticians. Studies measuring acoustic parameters of speech production have consistently shown that speakers apparently tailor their utterances on the basis of discourse structures, inserting longer pauses for major discourse boundaries than for minor discourse boundaries (den Ouden et al. 2009; Nakajima and Allen 1993; Noordman et al. 1999; Smith 2004; Swerts 1997; Yang and Yang 2012).

The present study

As was described above, a substantial body of literature has been devoted to the investigation of how readers integrate local information into a global discourse context to create a coherent representation. However, little is known about the effect of discourse structure on semantic integration. The question that we address is whether and how the difference of discourse structure affects semantic integration during on-line discourse comprehension. This is important because discourse structure is an integral part of discourse. Thus, accounts of discourse effects on semantic integration are incomplete if discourse structure is neglected. What we would like to propose is that, in the course of reading, readers construct not only a representation of discourse content, but also a representation of discourse structure. If readers capture both the content and the linguistic form in which the contents are conveyed, it should, in principle, be possible to observe a modulation effect of discourse structure on the integration of discourse content.

To test our hypothesis, we followed RST and constructed two kinds of discourse structures as described above (see discourses 1a and 1b and Fig. 1 for illustration). Semantic integration was manipulated by embedding a critical word in sentence 3, which was always semantically congruous in the local sentence context but was either congruous or incongruous when interpreted with sentence 2 of the discourse. Therefore, in the first kind of structure, semantic integration occurred within the same discourse unit, while in the second kind of structure, semantic integration occurred between two discourse units. The two factors of discourse structure and discourse congruence were crossed in a factorial design, which resulted in four conditions: incongruous, within unit; congruous, within unit; incongruous, between unit; congruous, between unit. We used commas to connect the three clauses in both structures, so that readers could infer the structures of the

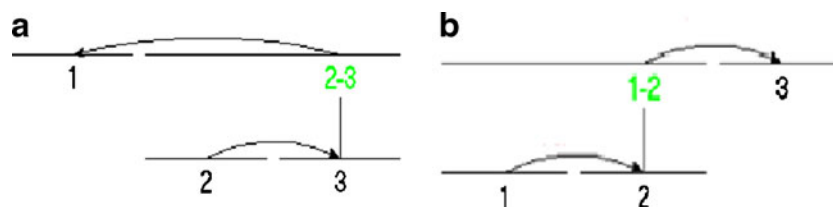


Fig. 1 Two kinds of rhetorical structure

discourses only from semantic information, but not syntactic markings. This is important since we did not want to confound the effects of discourse structure with the effects of syntactic markings. The use of commas in the materials may be ungrammatical in English. However, this is perfectly acceptable in Chinese. Unlike English commas, commas in Chinese play the same role as periods and are used frequently to connect complete sentences. Sentences that are connected only by commas are called run-on sentences in Chinese. For more information about the use of commas in Chinese segmentation, please see Xu et al. (2013).

Two experiments were conducted to investigate the effect of discourse structure on semantic integration. Experiment 1 used self-paced reading paradigm, while Experiment 2 used eye-tracking measures to further explore the time course of the effects. On the basis of previous studies showing that humans are very sensitive to the structural differences of language and that different response patterns could be found for these structural differences regardless of the techniques or measures used (Bahlmann et al. 2006; Bahlmann et al. 2008; Friederici 2004; Jiang and Zhou 2009; Makuuchi et al. 2009), we hypothesize that readers should be sensitive to the difference of discourse structure and a modulation effect of discourse structure on semantic integration should be observed. Since sentence 2 and sentence 3 were more closely related in the within-unit condition than in the between-unit condition, as indicated by RST (Mann and Thompson 1988) and reflected by shorter in-between pauses when speakers articulated the utterance (den Ouden et al. 2009; Nakajima and Allen 1993; Noordman et al. 1999; Smith 2004; Swerts 1997; Yang and Yang 2012), we expect that semantic anomalies should be detected more successfully in the within-unit condition than in the between-unit condition. Thus, the effect of discourse on local anomalies should be more pronounced in the within-unit condition than in the between-unit condition.

Experiment 1

In Experiment 1, we used a self-paced reading paradigm to examine the extent to which discourse structure affects semantic integration. This was measured in terms of the reading times for the regions of interest. On the basis of our initial discussion, we expected longer reading times for the incongruous words in the within-unit condition than in the between-unit condition.

Method

Participants

Twenty-six university students were recruited from China Agriculture University to participate in the experiment (14

women; mean age=21.0 years, $SD=2.6$). All were paid for participation.

Materials

Forty sets of three-sentence discourses were constructed (see Table 1 for example stimuli). The critical word always appeared in the middle of the last sentence, which was followed by another two words. Both the critical word and the following two words were two-character words. All of the words following the target word were identical across conditions. The incongruous and congruous words were matched on log frequency [incongruous, $M=3.67$, $SD=0.63$; congruous, $M=3.71$, $SD=0.76$; $t(39)=-0.39$, $p>.1$] on the basis of information provided by the Chinese Linguistic Data Consortium (2003). The mean numbers of words in the two kinds of discourses were also matched [within unit, $M=35$, $SD=2$; between unit, $M=35$, $SD=2$; $t(39)=-0.39$, $p>.1$]. Since semantic integration in the experimental material was manipulated to occur between the target word in the last sentence and the corresponding information in the preceding sentence, we also matched the integration distance so that the mean number of words that integration passed was also matched across the two structures [within unit, $M=9.12$, $SD=1.92$; between unit, $M=9.55$, $SD=2.09$; $t(39)=-1.19$, $p>.1$]. The 40 discourses were counterbalanced and divided into four lists, with each discourse set presented only once within each list. Each list contained 10 discourses per condition. To each list, 40 filler discourses of various structures were added, half of which involved discourse anomalies.

Three offline pretests of our experimental materials were conducted. In the first pretest, 30 participants were asked to rate the congruence of the discourses, ranging from 1 (*very implausible*) to 7 (*very plausible*). The participants were assigned one of the four counterbalanced lists. None of them participated in any other part of the experiment. Repeated measures ANOVAs revealed a main effect of congruence, $F(1, 29)=122.53$, $p<.001$, $\eta^2_{\text{partial}}=.81$ (mean $\pm$ $SD=2.66\pm 1.04$, 4.89 ± 1.11 , 2.55 ± 1.07 , and 4.83 ± 1.19 for within-unit/incongruent, within-unit/congruent, between-unit/incongruent, and between-unit/congruent, respectively). No main effect of discourse structure, $F(1, 29)=1.01$, $p>.1$, or interaction, $F(1, 29)=0.09$, $p>.1$, was found.

In the second pretest, 16 participants were asked to perform a cloze probability task. The experimental stimuli were truncated before the critical word and randomly intermixed with the 40 filler discourses. Two lists were created such that the participants could see only the within-unit or the between-unit condition of the same item, and they saw an equal number of discourses from the two conditions. The results again revealed that none of the participants completed the discourses with the incongruous words. Thus, the probability of an incongruous word was equivalently zero across the two structures. For the

Table 1 Example stimuli used in the present study**Within unit, incongruous (congruous)**

家人/都说/小李/准备/参加/运动会/,近来/他/总/去/田径场/,一遍遍地/努力/练习/
游泳/(跑步)/准备/比赛。/

The family/ says that /Li/ is going to /take part in /the sports meeting/,
lately /he /often/ went to /the track field/, over and over again/ tried /
practice/ **swimming**/ (**running**) /for/ the game/.

The family says that Li is going to take part in the sports meeting. Lately,
he's been going to the track field to practice **swimming** (**running**) over
and over again for the game.

Between unit, incongruous (congruous)

最近/小李/周末/总是/出门/,一遍遍地/在田径场/训练/,家人/都说/他/是去/练习/
游泳/(跑步)/ 备/比赛。/

Lately/ Li / on weekends /often /goes out/, over and over again/ in the
track field / training/, the family /says that /he/ went there to/ practice /
swimming / (**running**) /for/ the game/.

Lately, Li often goes out on weekends to train in the track field. His family
says that he goes there to practice **swimming** (**running**) for the game.

Note. Translations are presented below each Chinese sentence. Target words are marked in bold. The congruous target words are presented in parentheses.

congruous words, the cloze probability was 25.63 % ($SD = 12.63$ %) for the within-unit condition and 23.75 % ($SD = 13.10$ %) for the between-unit condition. A paired-samples t test by participants revealed no significant difference between the two kinds of structures, $t(15) = -0.35$, $p > .1$.

In the third pretest, 16 participants were asked to rate the coherence between the second sentence and the third sentence of the discourses on a scale of 1 (*not coherent*) to 5 (*very coherent*). Only the congruent conditions were presented, and the two conditions of discourse structure were crossed in a Latin square design with two sets of materials and two sets of participants. A paired-samples t test by participants showed that coherence scores were higher for the within-unit condition than for the between-unit condition [within unit, $M = 4.79$, $SD = 0.61$; between unit, $M = 3.67$, $SD = 0.63$; $t(13) = 5.12$, $p < .001$]. This result showed that both the within-unit and the between-unit structures were rated as coherent structures and that coherence scores were higher for the within-unit condition than for the between-unit condition. This finding is consistent with previous studies showing that minor discourse boundaries have subtler transitions and stronger semantic connections (den Ouden et al. 2009; Mann and Thompson 1988; Taboada and Mann 2006a, 2006b).

Procedure

The discourses were presented as white characters on a black background at the center of a computer screen, which was positioned approximately 70 cm away from the participants. Participants read the discourses in a noncumulative self-paced

reading paradigm. Each discourse began with a fixation cross presented at the middle of the screen for 1,000 ms. Then the first two sentences of the discourses appeared on the screen sentence by sentence, and the last sentence was presented one word at a time. Participants were instructed to press the space bar for the next sentence or word when they had read and understood the current sentence or word. To ensure that the participants read the discourses attentively, after each trial, they were asked to respond to a true/false comprehension question (e.g., “Is Li going to participate in the game?”). The participants responded to this question by pressing one of two appropriately marked keys. Half of the questions require a *true* response, and half a *false* response. After 10 practice items, the discourses were presented in a random order.

Results

Reading times less than 50 ms were excluded from analysis. We also excluded 2 participants with accuracy rates lower than 75 % in one of the conditions. Finally, outliers more than 2.5 standard deviations from the mean within a condition were excluded for each participant. Of the original data, 3.7 % was removed by these procedures. Mean reading times per region are displayed in Table 2. Data were reported for three regions: the target region consisting of the critical word, the posttarget region consisting of the word following the critical word, and the wrap-up region consisting of the final word of the sentence. We subjected data to analyses of variance (ANOVAs) with participants (F_1) and items (F_2) as random factors.

Target region

The main effect of discourse structure was only marginal in the analysis by participants, but it was not significant in the analysis by items, $F_1(1, 23) = 3.70$, $p = .07$, $\eta^2_{\text{partial}} = .14$; $F_2(1, 39) = 1.04$, $p > .1$, $\eta^2_{\text{partial}} = .03$. There was no significant interaction of discourse congruence and discourse structure ($F_s < 1$), nor was there a main effect of discourse congruence ($F_s < 2.8$).

Table 2 Mean reading times per region (in milliseconds) as a function of discourse congruence and discourse structure (with standard errors in parentheses)

	Within Unit		Between Unit	
	Incongruous	Congruous	Incongruous	Congruous
Target region	472 (34)	450 (34)	507 (39)	475 (26)
Posttarget region	508 (21)	463 (21)	504 (25)	501 (26)
Wrap-up region	507 (24)	477 (25)	498 (26)	474 (22)

Posttarget region

There was a marginal effect of congruence, $F_1(1, 23)=3.33$, $p=.08$, $\eta^2_{\text{partial}}=.13$; $F_2(1, 39)=3.65$, $p=.06$, $\eta^2_{\text{partial}}=.08$, with incongruent words read longer than congruent words. There was no significant main effect of discourse structure ($F_s<2.9$). However, there was an interaction between discourse congruence and discourse structure, $F_1(1, 23)=8.70$, $p<.01$, $\eta^2_{\text{partial}}=.28$; $F_2(1, 39)=3.51$, $p=.07$, $\eta^2_{\text{partial}}=.08$. Simple effects analyses revealed that in the within-unit condition, discourse-incongruous sentences were read 44 ms longer than discourse-congruous sentences, $F_1(1, 23)=8.43$, $p<.01$, $\eta^2_{\text{partial}}=.27$; $F_2(1, 39)=9.05$, $p<.01$, $\eta^2_{\text{partial}}=.19$, whereas in the between-unit condition, there was no significant difference between discourse-incongruous and discourse-congruous sentences ($F_s<1$). These results again provide evidence that semantic integration was influenced by discourse structure such that incongruous words in the within-unit condition were more elaborately processed.

Wrap-up region

The main effect of discourse congruence was only marginal in the analysis by participants, but not significant in the analysis by items, $F_1(1, 23)=5.82$, $p<.05$, $\eta^2_{\text{partial}}=.20$; $F_2(1, 39)=2.26$, $p>.1$, $\eta^2_{\text{partial}}=.05$. There was no significant interaction of discourse congruence and discourse structure ($F_s<1$), nor was there a main effect of discourse structure ($F_s<1$).

Discussion

ANOVAs indicated hints of processing disruption for discourse-incongruous words, as compared with discourse-congruous words, at the target region, suggesting that discourse-anomalous words were immediately related to the wider discourse context. More interestingly, at the posttarget region, discourse-incongruous sentences induced longer reading times than did discourse-congruous sentences only for the within-unit condition, but not for the between-unit condition. This provides clear evidence that semantic integration was affected by discourse structure. However, at the wrap-up region, there was a lingering main effect of discourse congruence in the analysis by participants. This could be a result of a discourse wrap-up effect, indicating that readers were engaged in building a coherent discourse model irrespective of discourse structure.

Note that the effect of discourse structure was found on the posttarget region, but not on the target region. This comes as no real surprise. The delayed effect is consistently found in the literature when a self-paced reading paradigm is used. Similar to the studies presented here, Hoeks et al. (2002) used a self-paced reading paradigm and eye tracking to examine the effect of topic-structure on the processing of ambiguous phrases. In their self-paced reading study, the modulation effect of topic-

structure on ambiguity processing was found not on the disambiguating verb, but one word later. However, when the same manipulation was tested using eye-tracking measures, the crucial interaction between context and semantic ambiguity and context type was already significant at the disambiguating verb itself, suggesting that eye tracking may be better able to capture early effects of structural difference.

Experiment 2

In Experiment 2, eye-tracking measures were used in order to examine the same factors of discourse congruence and discourse structure as in Experiment 1. This is important because eye tracking allows a normal presentation of language stimuli and provides a more time-sensitive measure of language processing. Note that in Experiment 1, the interaction between discourse congruence and discourse structure was found not on the target word, but on the posttarget words. Using eye movement measures, it is possible to obtain a more direct measure of early processing than self-paced reading.

Method

Participants

Twenty-eight undergraduates (14 males) participated for financial compensation. All had normal to corrected-to-normal vision. All were from the same population as in Experiment 1. None of them had participated in Experiment 1.

Materials

The materials of Experiment 1 were used.

Procedure

Eye movements were measured using EyeLink 1000, with a sampling rate of 1000 Hz. Participants read the stimuli with both eyes, but only the right eye was monitored. They were instructed to read normally for comprehension and were required to respond to a yes/no comprehension question after one fourth of the trials. Each trial began with a calibration check. Recalibration was performed during the experiment when the participants' fixations exhibited drift. Experimental session lasted 30–45 min.

Results

Performance on comprehension questions was high: They averaged 90 % correct, with no participants scoring below 80 %. Less than 2 % of the data was lost due to track losses (periods where the eyetracker lost the images of the eyes). Fixations

longer than 800 ms were trimmed to 800 ms (Camblin et al. 2007). Fixations less than 80 ms were combined with a fixation on an adjacent character, and those not next to another fixation were eliminated (Rayner 1998; Rayner et al. 2004).

Eye movements were then analyzed to provide the following measures: first-fixation duration (the duration of the first fixation in the region), gaze duration (the total time of all the fixations spent in a region before exiting to the left or right, provided that the eyes have not yet gone beyond the area of interest), go-past time (also called regression-path duration, the sum of all fixation durations from when the eyes first encounter a word to when the eyes leave the word to the right, including time spent rereading earlier material and time spent rereading the word itself), and total time (the sum of all fixation durations on a region of interest). All measures were calculated contingent on a region gaining a progressive first fixation during its first pass. That is, cases wherein the region of analysis was skipped during first-pass reading were excluded from final analysis (Camblin et al. 2007; Cunnings and Clahsen 2007; Van Gompel and Pickering 2001). This procedure excluded about 20 % of the data for the target region and 15 % of the data for the posttarget region from final analysis. Data were reported for two regions: the target region, consisting of the critical word, and the posttarget region, consisting of the word following the critical word. Since reading time measures pertaining to the final word of the sentence were unstable because it was skipped more than 50 % of the time, we did not report data for the wrap-up region. Due to program error, two conditions of one item and one condition of another item were missing in the eye-tracking program, and no data were recorded for those three conditions. Thus, we removed those two items to retain proper counterbalancing, leaving 38 items for later analysis.

Target region

Table 3 shows the means for the target region. The effect of discourse structure was only marginal in the participant analysis of first-fixation duration (9 ms), $F_1(1, 27)=3.38$, $p=.08$, $\eta^2_{\text{partial}}=.11$; $F_2(1, 37)=2.55$, $p=.12$, $\eta^2_{\text{partial}}=.06$. There was a significant effect of discourse congruence in total time (65 ms), $F_1(1, 27)=14.35$, $p<.01$, $\eta^2_{\text{partial}}=.35$; $F_2(1, 37)=10.34$, $p<.01$, $\eta^2_{\text{partial}}=.22$. The pattern was for reading times to be longer for incongruous words than for congruous words.

More important, there was a marginal interaction effect in total time, $F_1(1, 27)=3.81$, $p=.06$, $\eta^2_{\text{partial}}=.13$; $F_2(1, 37)=4.06$, $p=.05$, $\eta^2_{\text{partial}}=.10$. Simple effects analyses revealed that in the within-unit condition, discourse-incongruous sentences were read longer than discourse-congruous sentences (98 ms), $F_1(1, 27)=14.45$, $p<.01$, $\eta^2_{\text{partial}}=.35$; $F_2(1, 37)=13.68$, $p<.01$, $\eta^2_{\text{partial}}=.27$, whereas in the between-unit condition, there was no significant difference between discourse-incongruous and discourse-congruous sentences ($F_s<1.6$,

Table 3 Eye movement measures (with standard errors in parentheses) on the target region

Measures (ms)	Within Unit		Between Unit	
	Incongruous	Congruous	Incongruous	Congruous
First-fixation duration	238 (8)	230 (7)	244 (7)	242 (6)
Gaze duration	287 (12)	254 (8)	277 (12)	279 (10)
Go-past time	438 (35)	371 (23)	383 (36)	398 (23)
Total time	455 (27)	357 (17)	442 (25)	413 (18)

$p_s>.1$). The same interaction effect also reached significance in the participant analysis of gaze duration, $F_1(1, 27)=4.59$, $p<.05$, $\eta^2_{\text{partial}}=.15$; $F_2(1, 37)=2.03$, $p=.17$, $\eta^2_{\text{partial}}=.05$, and go-past time, $F_1(1, 27)=5.32$, $p<.05$, $\eta^2_{\text{partial}}=.17$; $F_2(1, 37)=1.33$, $p=.26$, $\eta^2_{\text{partial}}=.04$. These interaction effects were again due to the fact that reading times were longer for incongruous words than for congruous words in the within-unit condition [for gaze duration, 32 ms, $F_1(1, 27)=7.15$, $p<.05$, $\eta^2_{\text{partial}}=.21$; $F_2(1, 37)=4.17$, $p<.05$, $\eta^2_{\text{partial}}=.10$; for go-past time, 66 ms, $F_1(1, 27)=5.04$, $p<.05$, $\eta^2_{\text{partial}}=.16$; $F_2(1, 37)=1.05$, $p=.31$, $\eta^2_{\text{partial}}=.03$], but not in the between-unit condition ($F_s<1$, $p_s>.1$). No other effects were significant ($F_s<2.7$, $p_s>.1$).

Posttarget region

Table 4 shows the means for the posttarget region. The main effect of discourse structure was only marginal in the participant analysis of first-fixation duration (9 ms), $F_1(1, 27)=3.24$, $p=.08$, $\eta^2_{\text{partial}}=.11$; $F_2(1, 37)=0.78$, $p>.1$, $\eta^2_{\text{partial}}=.02$, and gaze duration (19 ms), $F_1(1, 27)=5.22$, $p<.05$, $\eta^2_{\text{partial}}=.16$; $F_2(1, 37)=1.87$, $p>.1$, $\eta^2_{\text{partial}}=.05$. There was a main effect of discourse congruence in the item analysis of go-past time (94 ms), $F_1(1, 27)=2.71$, $p>.1$, $\eta^2_{\text{partial}}=.09$; $F_2(1, 37)=9.23$, $p<.01$, $\eta^2_{\text{partial}}=.20$, with

Table 4 Eye movement measures (with standard errors in parentheses) on the posttarget region

Measures (ms)	Within Unit		Between Unit	
	Incongruous	Congruous	Incongruous	Congruous
First-fixation duration	257 (11)	238 (9)	259 (11)	256 (11)
Gaze duration	293 (15)	272 (14)	308 (16)	297 (15)
Go-past time	911 (88)	795 (85)	952 (77)	880 (76)
Total time	463 (24)	378 (22)	429 (22)	428 (23)

incongruous words read longer than congruous words. Analysis of total time showed a significant main effect of discourse congruence (43 ms), $F_1(1, 27)=10.11$, $p<.01$, $\eta^2_{\text{partial}}=.27$; $F_2(1, 37)=15.16$, $p<.001$, $\eta^2_{\text{partial}}=.29$. Again, the pattern was for reading times to be longer for incongruous sentences than for congruous sentences.

More important, a significant interaction was observed in total times, $F_1(1, 27)=6.35$, $p<.05$, $\eta^2_{\text{partial}}=.19$; $F_2(1, 37)=5.58$, $p<.05$, $\eta^2_{\text{partial}}=.13$. As with the interaction at the target region, simple effects analyses revealed that in the within-unit condition, discourse-incongruous sentences were read longer than discourse-congruous sentences (85 ms), $F_1(1, 27)=27.02$, $p<.001$, $\eta^2_{\text{partial}}=.50$; $F_2(1, 37)=15.78$, $p<.001$, $\eta^2_{\text{partial}}=.30$, whereas in the between-unit condition, there was no significant difference between discourse-incongruous and discourse-congruous sentences ($F_s<1$, $p_s>.1$). This interaction indicated that incongruous words were processed more successfully in the within-unit condition than in the between-unit condition. No other effects were significant ($F_s<2.7$, $p_s>.1$).

Discussion

The main findings of Experiment 1 using a self-paced paradigm were replicated in this experiment. Results indicated hints of earliest disruption on the target word in first fixation for discourse-incongruous words, as compared with discourse-congruous words. These hints of disruption became reliable disruption in total times and extended into the posttarget region, suggesting that incoming words were rapidly integrated into the global discourse context.

More important, we found a modulation effect of discourse structure on discourse congruence both on the target word and on the posttarget word. On the target word, there was some evidence for an interaction in the participant analysis of gaze duration and go-past time, which became statistically reliable in total times. The pattern was for the two within-unit conditions to differ, with longer reading times in the incongruous/within-unit than in the congruous/within-unit condition, but the two between-unit conditions did not differ. On the posttarget region, the crucial interaction was again reliably found in total times, indicating that the influence of discourse structure on discourse congruence also extended past the incongruous word. This pattern of results again provided clear evidence that semantic integration was affected by discourse structure such that the effect of discourse anomalies was more pronounced in the within-unit condition than in the between-unit condition.

In contrast to Experiment 1, the critical interaction between discourse structure and discourse congruence was already found at the target word itself, rather than at the posttarget word, indicating that eye tracking provides a more time-sensitive measure of processing than does self-paced reading.

General discussion and conclusion

The aim of the present study was to investigate whether and how the difference of discourse structure modulates semantic integration. To this end, we made use of discourse-dependent anomalies and manipulated semantic integration to occur within the same discourse unit or across two discourse units. Two key findings emerged: First, discourse-incongruous words were found to induce longer reading times than did discourse-congruous words; second and more important, discourse structure affected semantic integration such that the effect of discourse anomalies was more pronounced in the within-unit structure than in the between-unit structure.

The finding that discourse-dependent anomalies induce longer reading times is consistent with a strong effect of context. It is worth noting that the incongruous words in the present study were always congruous in the local sentence context and became incongruous only when interpreted with the discourse context. Therefore, the finding that the incongruous words induced reading disruption reinforces the fact that discourse context can override local lexical processing, which compares favorably with findings of previous studies (Camblin et al. 2007; Filik 2008; Filik and Leuthold 2008; Nieuwland and Van Berkum 2006; Van Berkum et al. 1999; Van Berkum et al. 2003).

The manipulation of discourse congruence involved discourse passages that were not highly constraining (around 25 % for the two kinds of structures as shown in cloze probability). However, our results showed an early and sustained effect of discourse congruence on the processing of incoming words. In Experiment 1, the main effect of discourse congruence (significant only by participants) emerged at the target region, which then became reliable at the posttarget region and extended into the wrap-up region. In Experiment 2, in which data for the wrap-up region were not reported, a marginal main effect of discourse congruence was already observed in first-fixation duration at the target region, which then became reliable in total times and extended into the posttarget region. These results are clearly at odds with the modular account of discourse processing in which new words are related to the discourse context after their meaning at the sentence level has been computed (e.g., Garrod and Sanford 1994). In contrast, our results are consistent with the one-stage model of discourse comprehension, which assumes that discourse-related information can take precedence over lexical processing of upcoming words in the discourse context (Van Berkum 2009).

Across the two experiments, modulation effects of discourse structure on semantic integration were found. In Experiment 1, we found that at the posttarget region, incongruous words induced longer reading times only in the within-unit structure, but not in the between-unit structure. In Experiment 2, this pattern was found at both the target region and posttarget region. This result adds to the literature by showing that

semantic integration is modulated by discourse structure. According to RST (Mann and Thompson 1988), successive sentences are more closely related when they share the same discourse purposes than when they are intended for different discourse purposes. This has been supported by acoustic studies showing that in production, speakers produce shorter in-between pauses when two successive sentences share the same discourse purposes than when they do not (den Ouden et al. 2009; Nakajima and Allen 1993; Noordman et al. 1999; Smith 2004; Swerts 1997; Yang and Yang 2012). In the present study, semantic integration was manipulated to take place between the third sentence containing the target word and the second sentence of the discourses. These two sentences shared the same discourse purposes in the within-unit condition, but not in the between-unit condition. Therefore, in the within-unit condition, the second and the third sentences were more closely related, and relevant information was more available. Thus, semantic anomalies were detected more easily, which resulted in longer reading times for anomalies, as compared with congruous words. In the between-unit condition, however, the second and the third sentences were only loosely related. Consequently, relevant information was less available. Therefore, semantic anomalies were not successfully detected.

Another possibility may be related to memory activation. It has been suggested that the more coherent the incoming information is with the previous information, the more likely it is to activate the same or connected memory nodes (Gernsbacher 1997). Therefore, it is possible that in the within-unit condition, relevant information is more highly activated. In this case, violation of the relevant information causes stronger conflict, which requires more thorough processing to resolve.

Note that the effect of discourse structure is distinct from the effect of semantic integration. Although both effects arise from discourse context, effects of semantic integration are about what has been said in the previous discourse context, while effects of discourse structure depend on the linguistic form in which the contents are conveyed. The present study, by manipulating both discourse structure and semantic integration, has provided evidence that readers construct both the representation of discourse structure and the representation of discourse context during online reading process. More important, we find evidence that the representation of discourse structure modulates the representation of discourse context. Specifically, readers construct a more detailed representation of discourse context when successive sentences are more closely related.

A related and somewhat perplexing issue concerns how and when discourse structure modulates semantic integration. In the present study, there was some evidence that effects of discourse congruence preceded the modulation effects of discourse structure. In Experiment 1, while the effect of discourse structure on semantic integration was found at the posttarget

word, a marginal main effect of discourse congruence was found at the target word. In Experiment 2, while the earliest hints for the modulation effect of discourse structure came out in gaze duration at the target word, a marginal main effect of discourse congruence was already observed in first-fixation duration. Thus, it appears that discourse congruence may have an earlier effect than the modulation effect of discourse structure. On the basis of these patterns, it is possible that at the initial stage of interpretation, readers recruit knowledge about discourse context to modify the default semantic features initially associated with a word. Then knowledge of discourse structure soon comes into play, when the initial interpretation is refined. At this stage, readers built a more detailed representation when successive sentences are more closely related than when they are only loosely related. However, due in part to problems with statistical power, this discussion of time course is only suggestive, and future studies are needed to further explore the timing of these discourse effects.

To summarize, the experiments reported in this study provided clear evidence that discourse structure has a robust impact on online semantic integration and that discourse effects arise not just from what the preceding sentences were about, but also from the way the preceding sentences were organized. Therefore, our findings have important implications for psycholinguistic accounts of discourse comprehension and suggest that theories of discourse comprehension should accommodate contributions of discourse structure to online discourse processing.

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Appendix 1 Experimental items

For every item, the within-unit condition is presented before the between-unit condition. The target words that varied in congruency are presented in parentheses, with the incongruous target words presented before the congruous target words.

1

人都说小李准备参加运动会,近来他总去田径场,一遍遍地努力练习(游泳/跑步)准备比赛。

近小李周末总是出门,一遍遍地在田径场训练,家人都说他是去练习(游泳/跑步)准备比赛。

The family says that Li is going to take part in the sports meeting, recently, he often went to the track field, and over and over again he practiced (swimming /running) for the game.

Recently, Li often went out on weekends, and kept training in the track field, the family says that he went there to practice (swimming /running) for the game.

2

张很喜欢逛超市,他经常去卖海鲜的柜台,精心挑选新鲜的(酸奶/海鱼)做成美食。

近小张经常去超市里,挑选新鲜的海鲜,我猜他很喜欢买(酸奶/海鱼)做成美食。

Zhang likes shopping in supermarkets, he frequently goes to the seafood counter, and carefully selects fresh (yoghourt /sea-fish) to make dinner.

Zhang frequently goes to the supermarket, and selects fresh seafood, I guess that he likes buying fresh (yoghourt /sea-fish) to make dinner.

3

雪很喜欢看书,她总去借科幻小说来看,然后和朋友探讨新看过的(言情/奇幻)小说情节。

雪总爱买书,还总买来科幻小说细读,大家都说她迷上了(言情/奇幻)小说情节。

Xiaoxue likes reading, she often borrows science fiction to read, and then discusses the newly-read (romantic /fantasy) novels with her friends.

Xiaoxue likes buying books, and she often buys science fiction to read, everybody says that Xiaoxue is obsessed with (romantic /fantasy) novels.

4

友都说小丽特别喜欢表演,她经常举办家庭舞会,为朋友表演(杂技/街舞)展示风采。

丽经常举办家庭聚会,为朋友表演舞蹈,朋友都说她喜欢表演(杂技/街舞)展示风采。

All her friends say that Xiaoli likes performing, she often hosts dancing parties, and performs (acrobatics /street dance) to show off to her friends.

Xiaoli often hosts family parties, and performs dancing for her friends, all her friends say that she likes performing (acrobatics /street dance) to show off.

5

诗养成了运动的习惯,她每晚都出去跑步,一吃完晚饭就换上衣服去(操房/操场)挥汗运动。

诗每天吃完晚饭后,换好了衣服就去跑步,同学都知道她每晚都去(操房/操场)挥汗运动。

Liushi has developed a habit of exercising, she goes out to run every night, she changes clothes immediately after she finishes dinner and goes to the (aerobic room /playground) for sports.

Every night after Liushi finishes dinner, she changes clothes and goes out to run, her classmates all know that every night she goes to the (aerobic room /playground) for sports.

6

人说小张身体不太好,最近她经常发烧,总是得上医院去接受(手术/输液)治疗才行。

张最近总是生病,动不动就发烧,家人说她经常得上医院去接受(手术/输液)治疗才行。

Her family says that Zhang is in poor health, she frequently has a fever, and she frequently goes to the hospital for (surgery /transfusion) treatment.

Zhang frequently falls ill, and frequently has a fever, her family says that Zhang always has to go to the hospital for (surgery /transfusion) treatment.

7

个公园非常有名,因为这里的小鸟很多,那些爱动物的人常来这里喂(小猫/鸽子)表达爱心。

个公园环境优美,还有很多小鸟,所以那些爱动物的人经常来这里喂(小猫/鸽子)表达爱心。

This park is very famous, because there are many birds in it, and people who love animals often come here to feed (cats /pigeon).

This park is very beautiful, and there are many birds in it, therefore people who love animals often come here to feed (cats /pigeon).

8

肖体形长得很胖,因为她经常光顾冰淇淋店,在那里连着吃几个(煎蛋/雪糕)以便解馋。

肖周末经常邀请同事,一起去光顾冰淇淋店,同事说她喜欢吃(煎蛋/雪糕)以便解馋。

Xiao looks fat, because she frequently goes to the ice cream store, and eats several (omelets /ice cream) to satisfy her craving.

On weekends Xiao often invites her colleague, and they go to the ice cream store together, her colleague says that Xiao likes eating (omelets /ice cream) to satisfy her craving.

9

家都觉得小陈做事认真,她经常一早就去图书馆,一整天都呆在那里(演讲/学习)充实自己。

陈经常一大早就起床,一整天都呆在图书馆复习,大家都觉得她很喜欢(演讲/学习)充实自己。

Everybody says that Chen works hard, she often goes to the library early in the morning, and for the whole day she will stay there (giving speeches /studying) to enrich herself.

Chen often gets up early in the morning, and for the whole day she stays in the library, everybody says that she likes (giving speeches /studying) to enrich herself.

10

家都说小李喜欢表演,他经常参加舞蹈比赛,为观众们表演自编的(相声/街舞)吸引眼球。

李经常参加比赛,为观众们展示自编的舞蹈,大家都说他喜欢表演(相声/街舞)吸引眼球。

Everybody says that Li loves performing, he often participates in dance competitions, and performs self-composed (cross talk /street dance) for the audience.

Li often takes part in competitions, and performs self-composed dancing for the audience, everybody says that Li loves performing (cross talk /street dance).

11

家都说小丽很喜欢购物,她经常去服装市场,一下子买很多(海鱼/裙子)满足自己。

丽经常去逛商场,为自己买很多衣服,大家都说她喜欢买(海鱼/裙子)满足自己。

Everybody says that Li likes shopping, she often goes to the clothes market, and buys lots of (seafood /skirts) to satisfy herself.

Li often goes to the shopping mall, and buys lots of clothes for herself, everybody say that she likes buying (seafood / skirts) to satisfy herself.

12

静是个外向的女孩,她经常去上辩论课,努力学习以便提高(气质/口才)提升自己。

静经常参加培训,努力学习各种辩论技巧,大家都说她想提高(气质/口才)提升自己。

Jing is an outgoing girl, she often attends the debate classes, where she works hard to improve her (grace /eloquence).

Jing often attends training lessons, where she works hard to learn the skill of debate, everybody says that she wants to improve her (grace /eloquence).

13

李很注重休闲,他总是去一家养生馆,在那里蒸几个小时(饺子/桑拿)放松身心。

李周末经常出门,而且总是去养生馆,大家都说他喜欢蒸(饺子/桑拿)放松身心。

Li pays great attention to recreation, he often goes to a health club, and for several hours he (steams dumplings /sits in a sauna) for relaxation.

Li often goes out on weekends, and goes to a health club, everybody says that he likes (steaming dumplings /sitting in a sauna) for relaxation.

14

家都说小张很有魅力,她经常去参加交谊舞会,一连跳好几遍(舞剧/恰恰)展示魅力。

张周末经常出门,参加各种交谊舞会,大家都说小张喜欢跳(舞剧/恰恰)展示魅力。

Everybody says that Zhang is a charming girl, she often goes to dance parties, and plays the (dance drama /dances the cha-cha) over and over again to show her charm.

Zhang often goes out on weekends, and attends dance parties, everybody says that Zhang likes playing the (dance drama /dancing the cha-cha) to show her charm.

15

萌对天文很感兴趣,她经常去观测月食,一晚上集中精神地观看(太阳/月亮)探索奥秘。

萌经常去天文台,和同伴一起观测月食,大家都说她喜欢观看(太阳/月亮)探索奥秘。

Mengmeng is interested in astronomy, she often goes to observe the eclipse, and for the whole night she keeps observing the (sun /moon) to explore its mysteries.

Mengmeng often goes to the observatory, and observes the eclipse with her friends, everybody says that she likes observing the (sun /moon) to explore its mysteries.

16

说小林是一个爱整理的人,他经常整理书柜,把里面不用的旧(鞋子/书籍)统统扔掉。

林一旦有空余时间,就会整理他的书柜,听说他喜欢把不用的旧(鞋子/书籍)统统扔掉。

Lin likes cleaning the house, he often tidies up the bookcase, and throws away all the old (shoes /books).

Whenever Lin has a few spare minutes, he tidies up the bookcase, it appears that he likes throwing away all the old (shoes /books).

17

家都说小明很有好奇心,他经常去海洋馆,整整一天兴奋地研究(花草/海豚)满足好奇。

明一旦有空余时间,就会去海洋馆,大家都说他很喜欢研究(花草/海豚)满足好奇。

Everybody says that Ming is very curious, he often goes to the aquarium, and for the whole day he studies (plants /dolphin) to satisfy his curiosity.

Whenever Ming has a few spare minutes, he goes to the aquarium, everybody says that he likes studying (plants /dolphin) to satisfy his curiosity.

18

友都说小赵喜欢收藏,他经常去古董市场,精心挑选很多珍贵的(盆栽/字画)以供欣赏。

赵一旦有空余时间,就会去逛古董市场,朋友都说他喜欢买珍贵的(盆栽/字画)以供欣赏。

His friends says that Zhao likes collecting, he often goes to the antique market, and carefully chooses rare (pottery / painting) for his own appreciation.

Whenever Zhao has spare time, he goes the antique market, his friends says that he likes buying rare (pottery /painting) for appreciation.

19

友都说小燕很摩登,她经常去逛服装店,购买各种最新潮的(手表/衣服)装饰自己。

燕经常跟朋友去逛商场,购买最新潮的服装,朋友都说她喜欢买时尚的(手表/衣服)装饰自己。

Her friends say that Yan looks fashionable, she often visits clothing stores, and buys various kinds of latest (watches / clothes) for herself.

Yan often visits shopping malls with friends, and buys fashionable clothes, her friends say that she likes buying latest (watches /clothes) for herself.

20

刘热衷学习,她每个周末都要上三个外语培训班,连续学习六个小时(绘画/英语)以便出国。

刘每周末都上三次培训班,连续学习六个小时外语,大家都说她学习(绘画/英语)以便出国。

Liu is eager to learn, every weekend she has three training classes for foreign languages, altogether she learns (painting / English) for six hours for the purpose of studying abroad.

Every weekend Liu has three training classes, and learns foreign language for six hours, everybody says that she is learning (painting /English) for the purpose of studying abroad.

21

刘很喜欢帮助别人,他经常去孤儿院,当场捐一大笔善款给(老人/孩子)表示关爱。

刘经常做慈善,捐好多钱给孤儿院,大家都说他乐于助人(老人/孩子)表示关爱。

Liu likes helping others, he often goes to the orphanage, and donates a large amount of money to the (old people /orphans).

Liu often does some charity, and donates lots of money to the orphanage, everybody says that he likes helping the (old people /orphans).

22

洪是个艺术爱好者,每当艺术厅展出钢琴家的作品,他都要跑去买梵高/(肖邦)精装作品。

当艺术厅有作品展出,小洪就跑去买钢琴家的作品,看来他喜欢买梵高/(肖邦)精装作品。

Hong is an art lover, whenever there are exhibitions of pianist works in the art hall, he goes there to buy works of (van Gogh /Chopin).

Whenever there are exhibitions in the art hall, Hong goes there to searching for works of pianists, it looks like he likes buying works of (van Gogh /Chopin).

23

冰很喜欢买东西吃,她经常去超市买零食,一下子买回很多(瘦肉/薯片)用来充饥。

冰经常去超市买东西吃,带回很多零食,估计她很喜欢吃(瘦肉/薯片)用来充饥。

Bing likes buying things to eat, she often goes to the supermarket for snacks, and buys lots of (meat /fried chips) to appease her hunger.

Bing often goes to the supermarket to buy things to eat, and brings back lots of snacks, it looks like she likes buying (meat /fried chips) to appease her hunger.

24

友都说莉莉很有气质,她经常买毛笔回家,努力地在家里练习(瑜伽/书法)修身养性。

莉莉经常去逛文具店,买回来很多毛笔,朋友说她喜欢练习 (瑜伽/书法)修身养性。

The friends say that Lily is a classy girl, she often buys writing brushes, and industriously practices (yoga /calligraphy) at home for self-cultivation.

Lily often goes to the stationary store, and buys many writing brushes, her friends say that she likes practicing (yoga /calligraphy) for self-cultivation.

25

家都说小张很爱生活,她经常和朋友去逛花市,一口气买回来很多(床单/鲜花)装饰房间。

张周末经常挤出时间,和朋友一起去逛花市,大家都说她喜欢买很多(床单/鲜花)装饰房间。

Everybody says that Zhang loves life, she often goes to the flower market with her friends, and buys lots of (sheets / flowers) to decorate the house.

Zhang often squeezes in time on weekends, and goes to the flower market with her friends, everybody says that she likes buying lots of (sheets /flowers) to decorate the house.

26

友都说小赵喜欢购物,她经常去服装专卖店,为自己买很多(鲜花/T恤)愉悦自己。

张经常去逛街,一下子买好几件衣服,大家都说她很喜欢买(鲜花/T恤)愉悦自己。

The friends say that Zhao likes shopping, she often goes to the clothing stores, and buys lots of (flowers /T-shirts) for herself.

Zhao often goes shopping, and buys lots of clothes, everybody says that she likes buying (flowers /T-shirts) for herself.

27

刘很喜欢出去玩,假期里他经常去海边,用相机拍摄(骆驼/海景)当做娱乐。

期里小刘经常出游,带上相机去海边,朋友说他喜欢拍摄(骆驼/海景)当做娱乐。

Liu likes going out, on vacations he often goes to the beach, where he uses his camera to take photos of (camels /the ocean-view) for fun.

On vacations Liu often goes out, and takes his camera to the beach, the friends say that he likes taking photos of (camels /the ocean-view) for fun.

28

木购物时很冲动,她经常出去逛书店,忍不住随手就买一套新的(衣服/小说)带回家里。

木经常周末出去,在书店里闲逛,听说她经常忍不住买新的(衣服/小说)带回家里。

Mumu buys things on impulse, she often goes to the bookstore, where she impulsively buys new (clothes /novels) and takes them home.

Mumu often goes out on weekends, and hangs out in the bookstore, I heard that she often impulsively buys new (clothes /novels) and takes them home.

29

林喜欢到处摄影,他经常去森林拍摄动物,专心地用相机拍摄(企鹅/黑熊)当做娱乐。

林假期经常出游,跑去森林拍摄野生动物,估计他喜欢拍摄(企鹅/黑熊)当做娱乐。

Lin loves photography, he often goes to the forest to take photos of animals, where he uses his camera to take photos of (penguins /black bears) for fun.

On vacations Lin often goes out, and goes to the forest to take photos of animals, it seems that he likes taking photos of (penguins /black bears) for fun.

30

明是很调皮的孩子,他上篮球课时经常故意捣乱,惹得(音乐/体育)老师生气。

明经常故意捣乱,弄得篮球课上不成,看来他喜欢惹得(音乐/体育)老师生气。

Ming is a naughty boy, he often makes mischief in basketball class, which annoys the (music /physics) teacher.

Ming often makes mischief, which interrupts the basketball class, it seems that he likes annoying the (music /physics) teacher.

31

陈学习英语很刻苦,他经常拿着词表,去操场很努力的背(课文/单词)以便出国。

陈每天都一早起床,去操场努力地背词表,大家都知道他背(课文/单词)以便出国。

Chen is an industrious student, he often takes his word list with him, and goes to the playground to recite the (texts / words) for studying abroad.

Chen often get up early in the morning, and goes to the playground to recite his word list, everybody knows that he is reciting the (texts /words) for studying abroad.

32

李很爱打扮,她经常去高跟鞋专卖店里,买时尚的(发卡/鞋子)搭配衣服。

李经常去专卖店里,买时尚的高跟鞋,看来她很爱买(发卡/鞋子)搭配衣服。

Li likes dressing up, she often goes to the shoe store, and buys fashionable (hairpins /shoes) to go with her clothes.

Li goes shopping, and buying fashionable shoes, it seems that she likes buying (hairpins /shoes) to go with her clothes.

33

张很喜欢有生活情调,她经常去花卉市场逛,买很多(杂草/鲜花)装饰房间。

张经常去逛市场,挑选各种花卉,家人都说她喜欢买(杂草/鲜花)装饰房间。

Zhang lives a quality life, she often goes to the flower market, and buys lots of (weeds /flowers).

Zhang often goes to the market, and buys various kinds of flowers, her family says that she likes buying (weeds /flowers).

34

学们都以为小陈喜欢艺术,他经常去画馆,在那里练习一下午(歌唱/绘画)提升自己。

经常去艺术馆,一下午都在那里画画,大家都说他热衷练习(歌唱/绘画)提升自己。

His classmates say that Chen loves art, he often goes to the painting house, and for the whole afternoon he practices (singing /painting) for self-cultivation.

Chen often goes to the art house, and for the whole afternoon he stays there painting, everybody says that he likes practicing (singing /painting) for self-cultivation.

35

丽很喜欢吃果蔬,她经常去卖水果的店里,一下子买几斤(萝卜/樱桃)补充营养。

丽经常去超市,一下子买几斤水果,我猜她很爱吃(萝卜/樱桃)补充营养。

Li likes eating fruits and vegetables, she often goes to the fruit store, and buys lots of (turnips /cherries) for nutritional supplement.

Li often goes to the supermarket, and buys lots of fruit, I guess that she likes eating (turnips /cherries) for nutritional supplement.

36

立很喜欢捕捉小动物,他经常去附近河边玩,给自己抓几只(麻雀/螃蟹)丰富生活。

立周末经常出门,去附近河边玩捉小动物,大家都说他喜欢捉(麻雀/螃蟹)丰富生活。

Li likes catching small animals, he often goes to the riverside nearby, where he catches a few (sparrows /crabs) for fun.

Li often goes out on weekends, and goes to the riverside nearby to catch small animals, everybody says that he likes catching (sparrows /crabs) for fun.

37

明是个勤劳的孩子,他经常拿拖把打扫卫生,很努力地把(书包/地板)擦得干净。

明经常一放学就回家,拿起拖把打扫卫生,听说他总是把(书包/地板)擦得干净。

Ming is laborious, he often uses the mop for house-keeping, and keeps cleaning the (schoolbag /floor) until it is spotless.

Ming often goes straight home after school, and uses the mop for housekeeping, I heard that he likes cleaning the (schoolbag /floor) until it is spotless.

38

张学习成绩很优秀,他经常参加数学竞赛,而且总能做对所有的(古诗/计算)题目拿奖。

校每次有竞赛,小张都会参加数学竞赛,听说他能做对所有的(古诗/计算)题目拿奖。

Zhang does well in his studies, he often takes part in math competitions, and correctly answers all of the (ancient poem / calculation) questions.

Whenever there is competition at school, Zhang takes part in the math competition, I heard that he can correctly answer all of the (ancient poem /calculation) questions.

39

肖很爱装饰家里,他经常去逛家电专卖店,购买漂亮的(地毯/灯饰)装饰客厅。

肖每次一拿到工资,就会去逛家电专卖店,估计他是去买(地毯/灯饰)装饰客厅。

Xiao likes decorating the house, he often goes to household Electrics stores, where he buys beautiful (carpets /lighting) to decorate the living room.

Every time Xiao gets his wages, he goes to the household Electrics stores, it seems that he goes there to buy (carpet / lighting) to decorate his living room.

40

家都说小吴喜欢饲养家禽,她经常去养鸡场看,一下子买回很多(鸭子/小鸡)放在院里。

吴经常去附近的养殖场,买回很多雏鸡,大家都说她喜欢买(鸭子/小鸡)放在院里。

Everybody says that Wu likes keeping poultry, she often goes to the chicken farm, and buys many (ducks /chickens) all at once to keep in her yard.

Wu often goes to the farm nearby, and buys many chicks, everybody says that she likes buying (ducks /chickens) to keep in the yard.

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